SC4001 – Part A: HDB Resale Price Regression

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Generative AI Declaration:

I used **Codex** to assist with the generation of selected code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

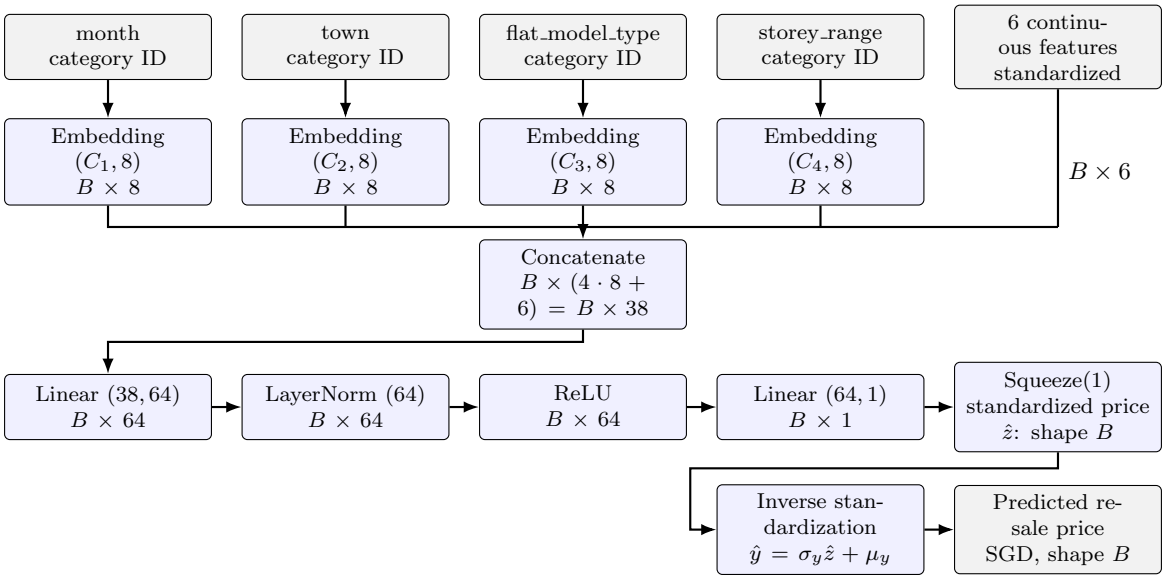
Generative AI Tool Used: Codex

Purpose of Use: Code generation and programming assistance

Verification Method: Manual code review, testing, debugging, and validation against assignment requirements

A1

(a) Block Diagram and Trainable Parameters



Dimensions: B is batch size. The baseline uses four 8-dimensional embeddings and six continuous features, giving $D = 4 \times 8 + 6 = 38$, with hidden width 64. Each C_j equals the number of known training categories plus one for unknown ID 0.

Categorical feature	Training categories	C_j
month	12	13
town	26	27
flat_model_type	43	44
storey_range	17	18

Continuous features: dist_to_nearest_stn, dist_to_dhoby, degree_centrality, eigenvector_centrality, remaining_lease_years, floor_area_sqm.

Component	Calculation	Parameters
Embeddings	$8(13 + 27 + 44 + 18)$	816
Linear $(38, 64)$	$38 \times 64 + 64$	2,496
LayerNorm (64)	$64 + 64$	128
Linear $(64, 1)$	$64 + 1$	65
Total	$816 + 2,496 + 128 + 65$	**3,505**

Linear layers include biases; LayerNorm includes learnable scale and shift. The remaining operations add no trainable parameters.

(b) Hyperparameter Search

Search: Ray Tune evaluated all 25 pairs of $h \in \{16, 32, 64, 128, 256\}$ and $d \in \{2, 4, 8, 12, 16\}$, using the same d for four categorical features. Training used 2017–2020; validation used 2021. Settings were seed 42, Adam, learning rate 0.01, batch size 512, maximum 20 epochs, and early stopping with patience 4 and minimum validation RMSE improvement 10^{-4}.

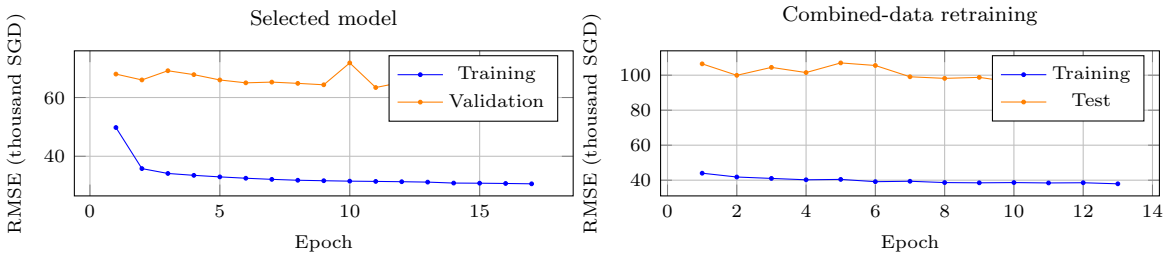
(c) Selected Configuration

Selection: The lowest best validation RMSE was **59,766.91 SGD**, achieved by $h = 128$, $d = 12$ at epoch 13. Training stopped at epoch 17 and epoch-13 weights were retained. The left plot below shows this model's training and validation curves; training RMSE aggregates batches during updates.

(d) Retraining and Test Performance

Retraining: A fresh selected model was trained on 116,427 records from 2017–2021, with preprocessing refitted on that set. Adam settings and seed 42 were retained. The duration was fixed at the validation-selected 13 epochs; test metrics never controlled training. An independent seed-42 shuffle generator matches subsequent experiments.

Final evaluation: On 26,702 test records from 2022, RMSE was **98,124.06 SGD** and $R^2 =$ **0.667858**; training RMSE was 37,926.73 SGD. The right plot shows end-of-epoch training/testing RMSE.



(e) Discussion

Capacity: The selected configuration improved validation RMSE over $(64, 8)$ (63,422.46 SGD) by 5.76%. The runner-up $(256, 2)$ scored 59,926.48 SGD, only 159.58 SGD higher. Larger capacity did not consistently help: at $h = 128$, $d = 16$ scored 65,599.97 SGD.

Fit and generalisation: From epochs 13 to 17, training RMSE declined from 31,207.16 to 30,627.12 SGD while validation RMSE rose from 59,766.91 to 69,319.56 SGD, indicating late overfitting and supporting early stopping. The final training/test gap suggests limited generalisation; temporal changes may contribute, so the gap alone cannot isolate overfitting. The curves do not suggest simple capacity-based underfitting.

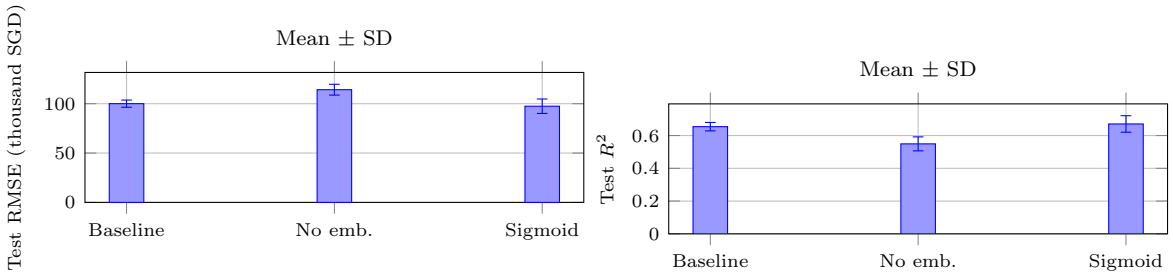
A2

Protocol: Baseline and both ablations were trained from scratch on 2017–2021 using seeds $\{42, 43, 44, 45, 46\}$, $h = 128$, $d = 12$ where applicable, and the A1 settings and 13 epochs. Preprocessing and minibatch orders were shared for matching seeds. Each final model was evaluated once on 2022. Results are mean ± sample SD ($n = 5$).

Per-seed results: RMSE in SGD.

Model	Seed	Train RMSE	Test RMSE	Test R^2
Baseline	42	37,926.73	98,124.06	0.6679
Baseline	43	38,258.40	97,431.21	0.6725
Baseline	44	37,931.61	101,068.42	0.6476
Baseline	45	37,476.46	97,601.34	0.6714
Baseline	46	38,618.59	106,030.81	0.6122
Without embeddings	42	54,406.94	114,211.19	0.5500
Without embeddings	43	55,308.87	117,697.22	0.5221
Without embeddings	44	55,730.65	119,940.38	0.5037
Without embeddings	45	53,933.65	113,674.84	0.5542
Without embeddings	46	54,487.32	105,507.78	0.6160
Sigmoid	42	38,004.36	96,090.96	0.6815
Sigmoid	43	39,493.07	109,142.47	0.5891
Sigmoid	44	38,728.61	89,514.55	0.7236
Sigmoid	45	36,920.54	98,480.10	0.6654
Sigmoid	46	38,093.72	94,102.55	0.6945

Model	Test RMSE (SGD)	Test R^2
Baseline	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256
Without embeddings	$114,206.28 \pm 5,500.29$	0.5492 ± 0.0427
Sigmoid	$97,466.13 \pm 7,310.91$	0.6708 ± 0.0504



(a) Removing Categorical Embeddings

Change: Remove all four embedding tables and change the first layer from $54 \to 128$ to $6 \to 128$, retaining LayerNorm, ReLU and output layer. Parameters decrease from 8,649 to 1,281.

Result: Mean test RMSE rose 14.15%; mean R^2 fell from 0.6543 to 0.5492. Four of five paired seeds worsened. Mean training RMSE rose from 38,042.36 to 54,773.49 SGD, supporting the value of categorical information; information loss and reduced parameter count are confounded.

(b) Replacing ReLU with Sigmoid

Change: Replace only the hidden activation, retaining all inputs and 8,649 parameters.

Result: Mean test RMSE fell 2.58%, but SD increased from 3,652.75 to 7,310.91 SGD; three of five paired seeds improved. Mean training RMSE was similar (38,248.06 versus 38,042.36 SGD). The bounded activation changes representations and gradients; the modest mean improvement and larger variation do not establish a reliable advantage over ReLU.

A3

(a) Selecting λ

Architecture: The linear and deep branches share the 54-dimensional input. The deep branch retains the A1 MLP ($h = 128$, $d = 12$); the wide branch is $54 \to 1$. Outputs combine as $\text{MLP}(x) + \lambda\text{Linear}(x)$ before inverse target scaling.

Search: Five coefficients were trained with seed 42, 2017–2020 training and 2021 validation, using A1's Adam and early-stopping settings. All candidates shared initial weights and independently seeded minibatch orders. Selection used the lowest best validation RMSE; the test set was unused.

λ	Best validation RMSE (SGD)	Best epoch	Epochs run
0	59,720.34	4	8
0.25	60,935.69	4	8
0.5	62,760.50	4	8
1	62,025.90	6	10
2	63,258.46	6	10

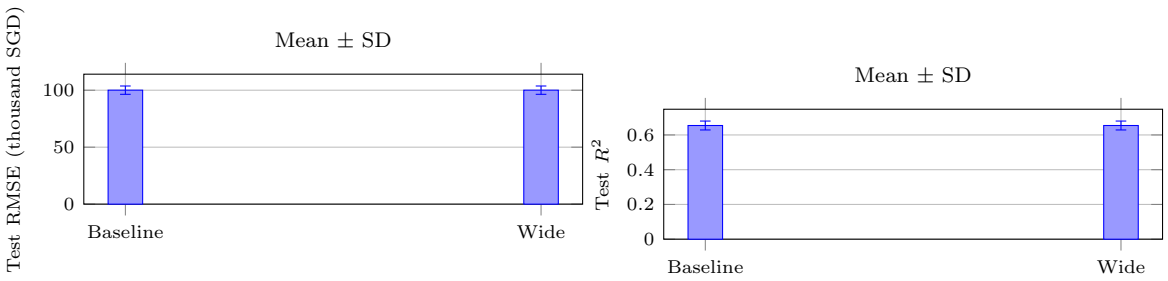
Selection: $\lambda = 0$ achieved 59,720.34 SGD at epoch 4; training stopped at epoch 8. All positive coefficients performed worse. The shuffle generator differs from the original A1 validation search, so its zero-coefficient result need not reproduce A1's 59,766.91 SGD.

(b) Five-Seed Evaluation

Protocol: The selected $\lambda = 0$ model was retrained on 2017–2021 with seeds 42–46 and A1's fixed 13 epochs, preprocessing and optimiser settings. Each final model was tested once. The A2 five-seed A1-architecture baseline supplies the matched comparison; its seed-42 weights equal the final A1(d) weights.

Seed	Train RMSE (SGD)	Test RMSE (SGD)	Test R^2
42	37,926.73	98,124.06	0.6679
43	38,258.40	97,431.21	0.6725
44	37,931.61	101,068.42	0.6476
45	37,476.46	97,601.34	0.6714
46	38,618.59	106,030.81	0.6122

Model	Test RMSE (SGD)	Test R^2
Baseline	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256
Wide-and-deep	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256



Comparison: All five paired metrics and embedding/MLP weights matched exactly because $\lambda = 0$ disables the wide contribution. No improvement was observed. A1(d)'s seed-42 reference is 98,124.06 SGD and $R^2 = 0.667858$; the five-seed mean accounts for run-to-run variation.

(c) Discussion

Mechanism and evidence: The linear branch offers a shorter gradient path to shared embeddings and may learn additive patterns alongside nonlinear MLP relationships. All positive coefficients had worse validation RMSE, providing no evidence of a benefit under these settings.

Limitations: Gradient norms were not measured, so relief of vanishing gradients cannot be verified. Coefficient selection used one seed, and final testing with $\lambda = 0$ cannot establish how positive coefficients generalise.

A4

(a) FeatureAblation Computation

Models: Load the actual final A1(d) baseline and A3(b) wide model, both trained with seed 42, matching shuffle order, preprocessing and 13 epochs on 2017–2021. Captum FeatureAblation was applied to all 26,702 test samples without retraining.

References: Categories use combined-training modes: month 7, town SENGKANG, flat model "4 ROOM, Model A", storey range "04 TO 06". Continuous features use zero after standardisation (combined-training means). Each categorical ID represents one entire embedding; the six continuous features are separate groups.

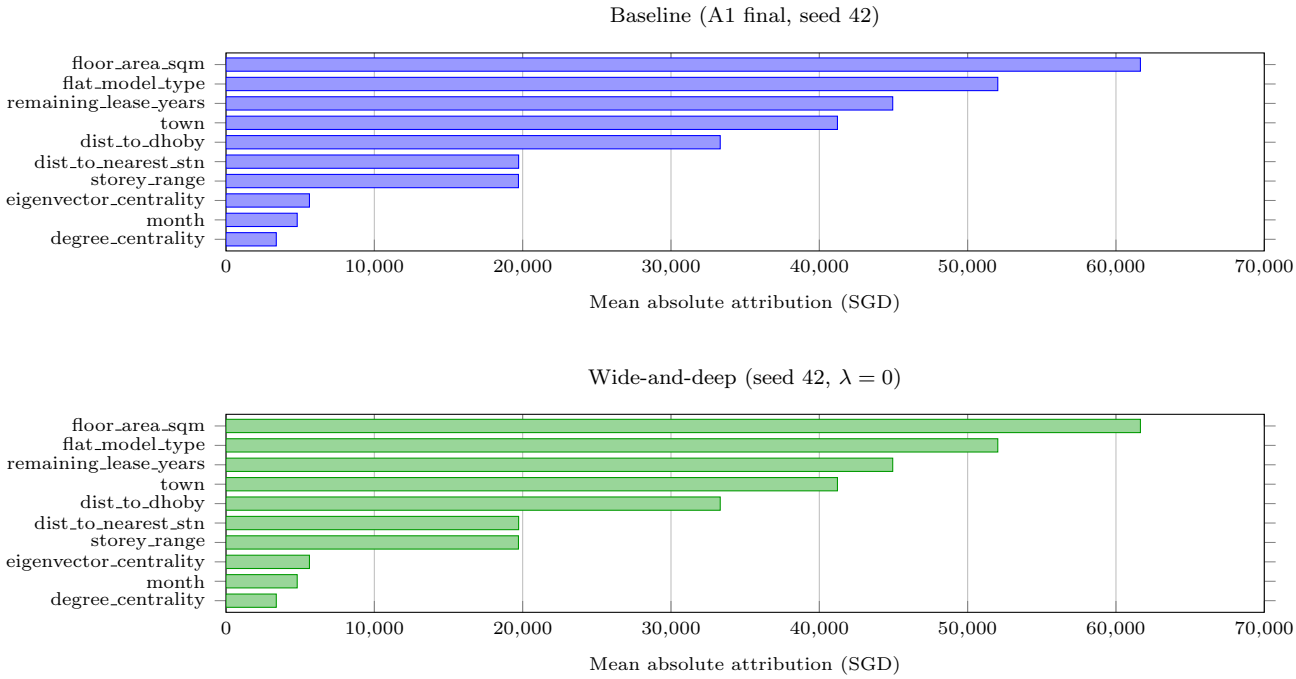
Aggregation: Predictions and attributions are in SGD. For feature reference r_j,

$$a_{ij}^{(m)} = f_m(x_i) - f_m(x_i^{(j \leftarrow r_j)}), \qquad I_j^{(m)} = \frac{1}{26{,}702} \sum_{i=1}^{26{,}702} |a_{ij}^{(m)}|.$$

Original feature	Baseline (SGD)	Wide-and-deep (SGD)
month	4,800.15	4,800.15
town	41,222.37	41,222.37
flat_model_type	52,032.36	52,032.36
storey_range	19,716.84	19,716.84
dist_to_nearest_stn	19,727.62	19,727.62
dist_to_dhoby	33,319.83	33,319.83
degree_centrality	3,389.72	3,389.72
eigenvector_centrality	5,625.70	5,625.70
remaining_lease_years	44,949.20	44,949.20
floor_area_sqm	61,648.57	61,648.57

Verification: Manual prediction differences matched Captum for ten features on eight samples. Both saved attribution matrices have shape $26{,}702 \times 10$, finite values and maximum inter-model difference 0 SGD.

(b) Feature Importance Plots



(c) Interpretation

Top three: Both models rank floor_area_sqm (61,648.57 SGD), flat_model_type (52,032.36 SGD) and remaining_lease_years (44,949.20 SGD) highest. These scores measure average prediction changes relative to the chosen references, rather than unit-price effects.

Comparison: Area, flat type and lease duration carry substantial predictive information. Both complete rankings and scores coincide because the selected $\lambda = 0$ model has the same predictive function as the matched baseline.

Limitation: Reference choices can change feature rankings, and replacing one feature may produce unrealistic combinations of correlated inputs. The resulting importance is not a causal effect.